

RETAIL ANALYTICS • FINAL REPORT

Customer Shopping Behavior Analysis

A data-driven study of spending patterns, customer segments, and subscription behavior across 3,900 retail transactions

Prepared by: Customer Analytics Team
Methods: Python (pandas) · PostgreSQL (SQL) · Power BI

1. Executive Summary

This report analyzes customer shopping behavior using transactional data from 3,900 purchases spanning four product categories: Clothing, Accessories, Footwear, and Outerwear. The dataset captures customer demographics, purchase details, and shopping behavior across 18 columns. The analysis followed a three-stage workflow — data preparation in Python, structured querying in PostgreSQL, and visualization in Power BI — to answer ten targeted business questions and translate the findings into strategic recommendations.

Metric	Value
Transactions analyzed	3,900
Data columns	18
Average purchase amount	\$59.76
Average review rating	3.75 / 5
Subscribed customers	1,053 (27%)
Missing values imputed	37 (Review Rating)

Headline Finding

Male customers generate over twice the revenue of female customers, average spend is nearly identical across subscription status, and revenue is evenly distributed across age groups — suggesting growth opportunities lie less in demographic targeting and more in converting repeat buyers into subscribers.

2. Dataset Summary

The dataset consists of 3,900 rows and 18 columns, organized into three thematic groups:

- **Customer demographics:** Age, Gender, Location, Subscription Status
- **Purchase details:** Item Purchased, Category, Purchase Amount, Season, Size, Color
- **Shopping behavior:** Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

The Review Rating column contained 37 missing values, which were imputed using the median rating of each product category to preserve category-level rating patterns without introducing bias from a global average.

3. Methodology

3.1 Data Preparation in Python

1. **Data Loading** — Imported the raw dataset using pandas.
2. **Initial Exploration** — Used `df.info()` and `.describe()` to check structure and summary statistics.
3. **Missing Data Handling** — Checked for null values; imputed Review Rating using the median rating per product category.
4. **Column Standardization** — Renamed columns to snake_case for readability and documentation.
5. **Feature Engineering** — Created `age_group` (binned ages) and `purchase_frequency_days` columns.
6. **Consistency Check** — Verified `discount_applied` vs. `promo_code_used` redundancy; dropped `promo_code_used`.
7. **Database Integration** — Loaded the cleaned DataFrame into PostgreSQL for SQL-based analysis.

3.2 SQL Analysis in PostgreSQL

Ten structured business questions were answered using SQL queries against the cleaned dataset, covering revenue breakdowns, discount behavior, product ratings, shipping comparisons, subscription analysis, customer segmentation, and category-level performance. Results are detailed in Section 4.

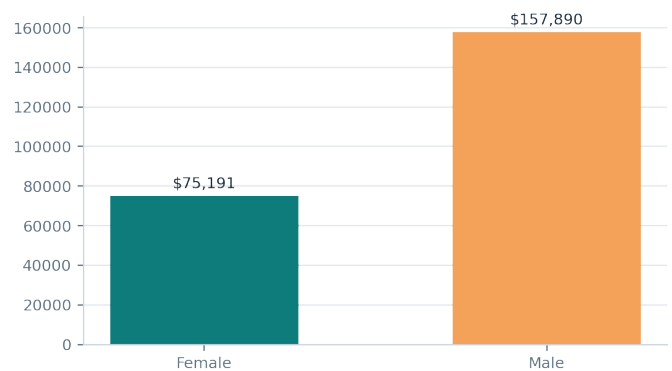
3.3 Visualization in Power BI

An interactive Power BI dashboard was built on top of the cleaned dataset, featuring filters for Subscription Status, Gender, Category, and Shipping Type, alongside summary KPIs and category/age-group breakdowns. See Section 5.

4. Key Findings

4.1 Revenue by Gender

Total revenue generated by male vs. female customers.



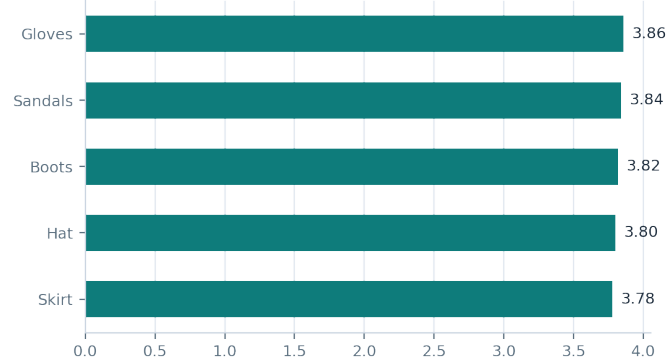
Gender	Revenue (USD)
Female	\$75,191
Male	\$157,890

Male customers generate more than double the revenue of female customers.

4.2 High-Spending Discount Users

839 customers used a discount or promo code yet still spent above the average purchase amount of \$59.76 — indicating discounts attract a segment of generous spenders, not only price-sensitive shoppers.

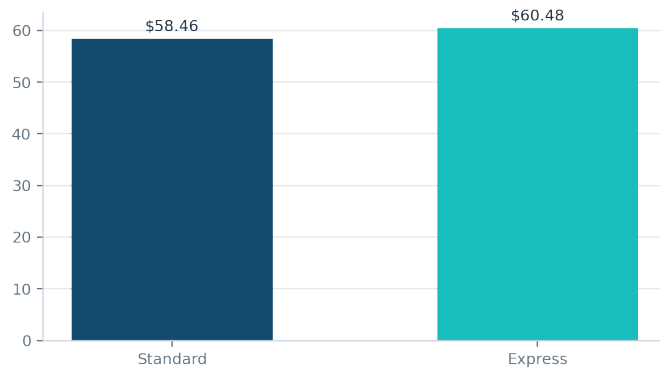
4.3 Top 5 Products by Average Rating



Product	Avg. Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

4.4 Shipping Type Comparison

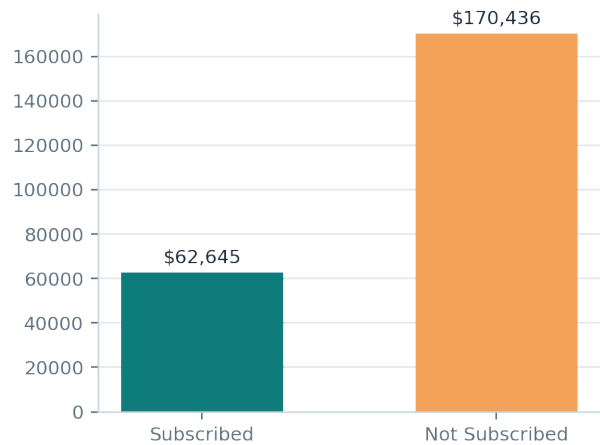
Average purchase amounts between Standard and Express shipping.



Shipping Type	Avg. Purchase
Standard	\$58.46
Express	\$60.48

Express shipping customers spend only marginally more on average.

4.5 Subscribers vs. Non-Subscribers

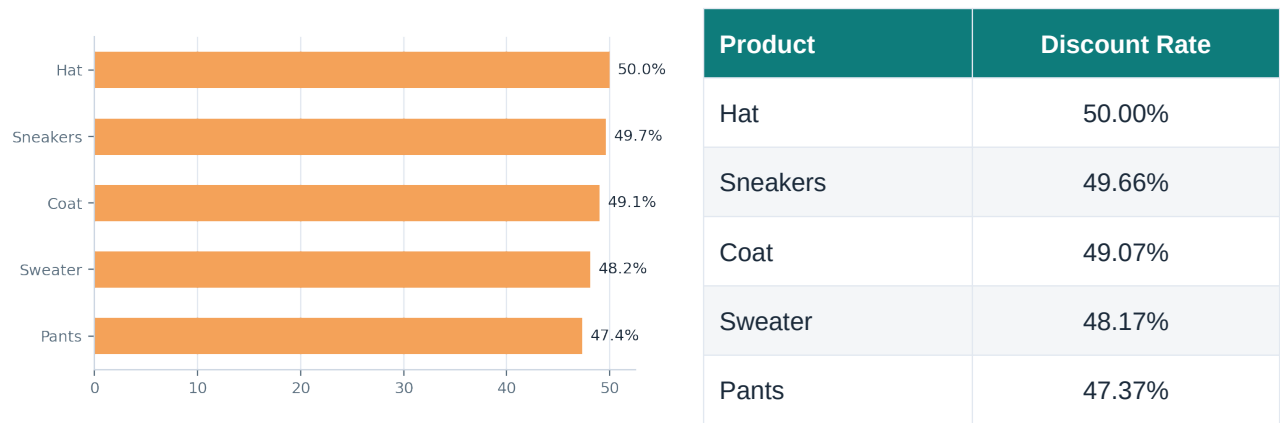


Subscription	Customers	Avg. Spend	Total Revenue
Yes	1,053	\$59.49	\$62,645
No	2,847	\$59.87	\$170,436

Average spend is nearly identical between subscribers and non-subscribers; non-subscribers drive more total revenue simply due to greater numbers.

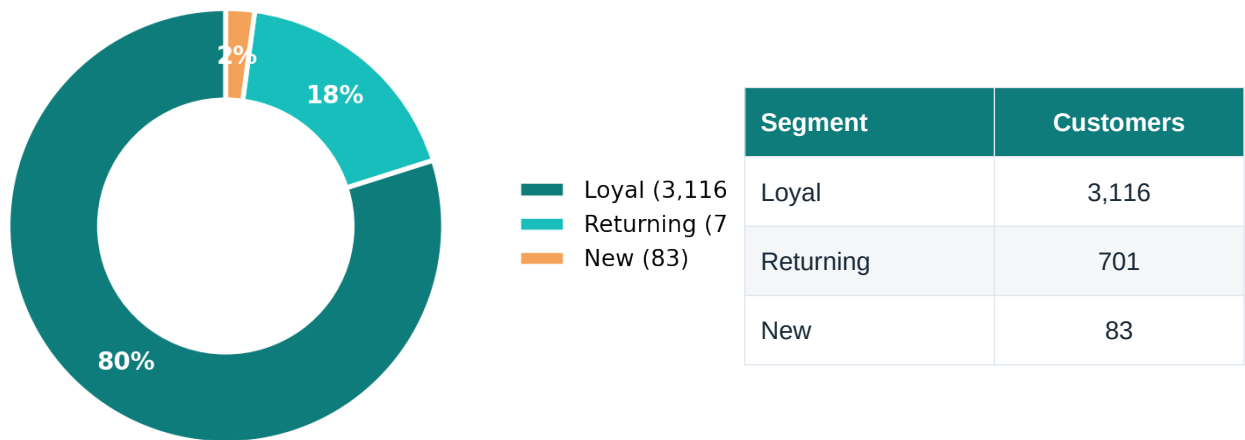
4.6 Discount-Dependent Products

Products with the highest percentage of discounted purchases.



4.7 Customer Segmentation

Customers classified into New, Returning, and Loyal segments based on purchase history.



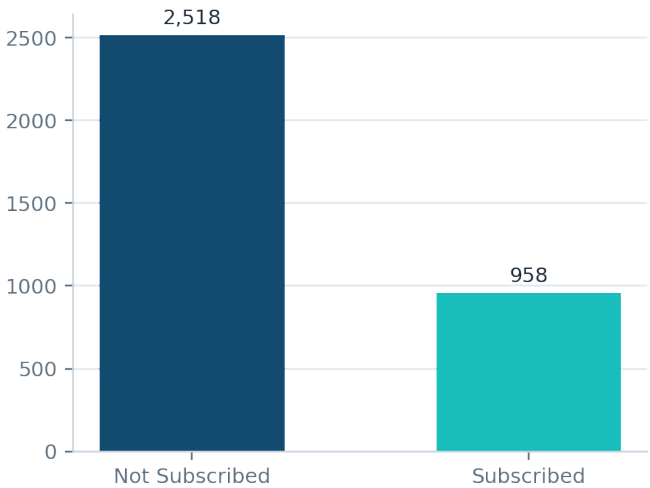
The vast majority of customers (80%) already fall into the Loyal segment.

4.8 Top-Selling Product per Category

Category	Top Product	Total Orders
Accessories	Jewelry	171
Clothing	Blouse	171
Footwear	Sandals	160
Outerwear	Jacket	163

4.9 Repeat Buyers & Subscriptions

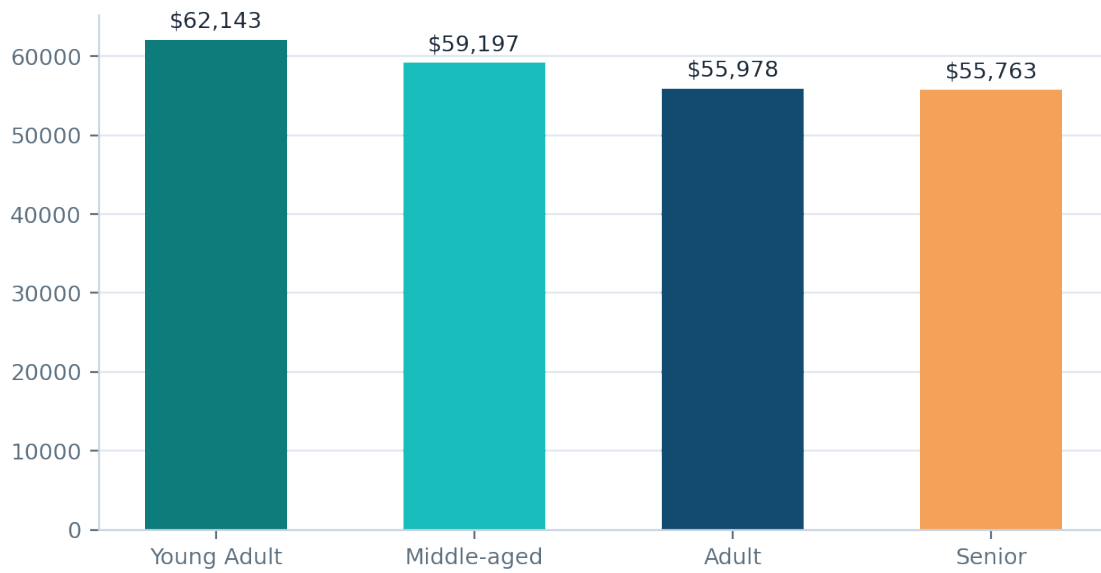
Whether customers with more than 5 purchases are more likely to subscribe.



Subscription	Repeat Buyers
No	2,518
Yes	958

Most repeat buyers (2,518 of 3,476) are still not subscribed — a clear conversion opportunity.

4.10 Revenue by Age Group



Age Group	Total Revenue
Young Adult	\$62,143
Middle-aged	\$59,197
Adult	\$55,978
Senior	\$55,763

Key Takeaway

Revenue is remarkably balanced across age groups — the spread from top to bottom is under 11%. Broad, age-inclusive marketing may outperform narrowly age-targeted campaigns.

5. Interactive Dashboard in Power BI

The SQL findings were translated into an interactive Power BI dashboard, enabling stakeholders to explore the data dynamically. The dashboard includes filters for Subscription Status, Gender, Category, and Shipping Type, alongside headline KPIs and category/age-group breakdowns of revenue and sales volume.

KPI	Value
Number of Customers	3.9K
Average Purchase Amount	\$59.76
Average Review Rating	3.75
Subscribed Customers	27%

Additional dashboard panels include Revenue by Category, Sales by Category, Revenue by Age Group, and Sales by Age Group — each filterable in real time.

6. Business Recommendations

1. Boost Subscriptions

Promote exclusive benefits for subscribers — only 27% of customers currently subscribe, despite comparable average spend to non-subscribers.

2. Customer Loyalty Programs

Reward repeat buyers to move them from the Returning segment into Loyal, reinforcing habitual purchasing.

3. Review Discount Policy

Balance sales boosts with margin control, particularly for heavily discounted products such as Hats and Sneakers.

4. Product Positioning

Highlight top-rated and best-selling products — Gloves, Sandals, and Boots — in marketing campaigns.

5. Targeted Marketing

Focus efforts on high-revenue age groups (Young Adults) and Express-shipping users, while maintaining broad age-inclusive reach given the balanced revenue spread.

7. Conclusion

This analysis of 3,900 customer transactions reveals a retail base with balanced revenue contribution across age groups, a large already-loyal customer segment, and an underexploited opportunity to convert repeat buyers into subscribers. Discounting is widespread but does not solely attract low-spend customers, suggesting the current discount policy could be refined for margin without sacrificing volume. Together, these findings support a strategy centered on subscription growth, loyalty rewards, and disciplined discount management rather than heavy demographic segmentation.